

Apply filters to SQL queries

Project description

The security team at my organization has found evidence of security issues and wants to strengthen and secure the internal systems for all employees. It is my job to ensure the system is safe, to investigate all potential security issues, and update employee computers as needed.

The following project details the steps I used to accomplish these tasks.

Retrieve after hours failed login attempts

There was a potential security incident that occurred after normal business hours – after 6:00pm. Protocol dictates all failed log-in attempts must be investigated.

The following code demonstrates how I created a SQL query to filter for failed login attempts that occurred after business hours. The first part of the screenshot is my query and the second part is a portion of the output.

```
MariaDB [organization]> clear
MariaDB [organization]> select *
  -> from log_in_attempts
  -> where login_time > '18:00' and success = 0;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0

Query entered:

```
SELECT *
FROM log_in_attempts
WHERE login_time > '18:00' AND success = 0;
```

I wanted to access all data from the `log_in_attempts` table and used the `WHERE` clause with an `AND` operator to create two conditions for the output to report: condition #1 `login_time > '18:00'` to only report login attempts after 18:00 and condition #2 `success = 0` to only report results where success 1 for true and 0 for false.

The output reported 19 reported failed log-in attempts after 6:00pm (18:00).

Retrieve login attempts on specific dates

A suspicious event occurred on May 9, 2022. Protocol dictates an investigation into all login attempts on the day in question and the day prior must be conducted.

The following code demonstrates how I created a SQL query to filter all login attempts that occurred on May 8 and May 9, 2022. The first part of the screenshot is my query and the second part is a portion of the output.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1

Query entered:

```
SELECT *  
FROM log_in_attempts  
WHERE login_date = '2022-05-09' or login_date = '2022-05-08';
```

I wanted to access all data from the `log_in_attempts` table and used the `WHERE` clause with an `OR` operator to create an output of either of the two conditions: condition #1 `login_date = '2022-05-09'` to only report login attempts on May 9, 2022 and condition #2 `login_date = '2022-05-08'` to only report login attempts on May 8, 2022.

There were 75 reported attempted log-ins on the dates of May 8 and May 9, 2022.

Retrieve login attempts outside of Mexico

Initial investigations with the outputs from the report data suggest that the suspicious activity on login attempts did not originate within the country of Mexico. To determine the origin, I continued to investigate all login attempts that occurred outside of Mexico.

The following code demonstrates how I created a SQL query to filter all login attempts occurring outside of Mexico. The first part of the screenshot is my query and the second part is a portion of the output.

```
MariaDB [organization]> select *  
-> from log_in_attempts  
-> where not country like 'Mex%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0

Query entered:

```
SELECT *  
FROM log_in_attempts  
WHERE NOT country LIKE 'Mex%';
```

I wanted to access all data from the `log_in_attempts` table and used the `WHERE` clause with an `NOT` operator to create a condition for my query to filter out login attempts originating in Mexico. I included a second condition using the `LIKE` operator with `'Mex%'` since the dataset represents Mexico as Mex.

There were 144 log-in attempts from the United States of America and Canada. .

Retrieve employees in Marketing

The security team wants to perform essential security updates on the machines for the employees in the east wing, specifically within the marketing department. Since I am responsible for patching these updates, I have to get access to each employee's information and the locations of machines they are using.

The following code demonstrates how I created a SQL query to filter for employee machines from employees in the marketing department in the east wing. The first part of the screenshot is my query and the second part is a portion of the output.

```
MariaDB [organization]> SELECT *  
  -> FROM employees  
  -> WHERE department = 'Marketing' AND office LIKE 'East%';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
|          1000 | a320b137c219 | elarson | Marketing | East-170 |  
|          1052 | a192b174c940 | jdarosa | Marketing | East-195 |  
|          1075 | x573y883z772 | fbautist | Marketing | East-267 |  
|          1088 | k865l965m233 | rgosh | Marketing | East-157 |  
|          1103 | NULL | randers | Marketing | East-460 |  
|          1156 | a184b775c707 | dellery | Marketing | East-417 |  
|          1163 | h679i515j339 | cwilliam | Marketing | East-216 |  
+-----+-----+-----+-----+-----+  
7 rows in set (0.001 sec)
```

Query entered:

```
SELECT *  
FROM employees  
WHERE department = 'Marketing' AND office LIKE 'East%';
```

I wanted to access all data from the `employees` table and used the `WHERE` clause with an `AND` operator to report on both conditions of my query: condition #1 `department = 'Marketing'` to only report employees in the marketing department and condition #2 `office LIKE 'East%'` to only report offices in the East wing.

The output of the query needed to satisfy both conditions 1 and 2, resulting in only 7 reported users in the marketing department who will need these security updates patched into their machines.

Retrieve employees in Finance or Sales

The security team wants to also perform separate security updates on the machines for the employees in the finance and sales departments, regardless of office location.

The following code demonstrates how I created a SQL query to filter for employee machines from employees in both the finance and sales marketing departments. The first part of the screenshot is my query and the second part is a portion of the output.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109

Query entered:

```
SELECT *  
FROM employees  
WHERE department = 'Finance' OR department = 'Sales';
```

I wanted to access all data from the `employees` table and used the `WHERE` clause with an `OR` operator to report on either one of the two conditions of my query: condition #1 `department = 'Finance'` to report employees in the finance department and condition #2 `department = 'Sales'` to report employees in the sales department.

The output of the query needed to satisfy only one of the two conditions resulting in 71 reported users in the in both the finance and sales departments who will need these security updates patched into their machines.

Retrieve all employees not in IT

The security team wants to also perform one last set of security updates for the employees of the organization. The IT team already has these updates installed, so it is crucial to find all other employees within the organization to see what machines will need these updates.

The following code demonstrates how I created a SQL query to filter for employee machines not in the IT department. The first part of the screenshot is my query and the second part is a portion of the output.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127

Query entered:

```
SELECT *  
FROM employees  
WHERE NOT department = 'Information Technology';
```

I wanted to access all data from the `employees` table and used the `WHERE` clause with a `NOT` operator to only report employees outside of the IT department under the conditional clause `NOT department = 'Information Technology'`.

The output of the query only needed to satisfy one condition resulting in 161 reported users across all other departments who will need these final security updates patched into their machines.

Summary

I applied different filters to SQL queries to get specific information on login attempts following suspicious activity and events and to find information on employee machines.

In this project, I searched for information within two tables – `log_in_attempts` and `employees`.

I used the `AND`, `OR`, and `NOT` operators to filter for the specific information needed for each task along with the `LIKE` and the `%` wildcards to filter for patterns within datasets.